

Prajwal Koirala

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[github/PrajwalKoirala](https://github.com/PrajwalKoirala)

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[Google Scholar](#)

SUMMARY

Graduate student with a background in robot perception, planning, and control, demonstrating a dedication to cutting-edge research in autonomous systems and reinforcement learning.

EDUCATION

Cornell University, Ithaca, New York (Aug 2025 - Present)

Ph.D. in Robotics, **GPA: 4.16**

Minors: Mechanical Engineering and Computer Science

Iowa State University, Ames, Iowa (Aug 2023 - Aug 2025)

M.S. in Mechanical Engineering, **GPA: 4.0**

Thesis: *Towards Safe and Efficient Offline Reinforcement Learning: Learning Safety Constraints and Expressive Policies via Generative Modeling*

Key Courses: (AERE, COMS, EE, ME)

- Machine Learning and Deep Learning
- Random Signals and Kalman Filters
- Linear Systems
- Computational Methods II (Data Str. & Algorithms)
- Convex Optimization
- Machine Vision

IOE Pulchowk Campus, Tribhuvan University, Lalitpur, Nepal (Nov 2017 - April 2022)

B.E. in Mechanical Engineering, **Score: 80.14%**

RESEARCH

Research Assistant, Autonomous Systems Lab, Cornell University (Aug 2025 – Present)

Advisor: Prof. Mark Campbell

- Autonomous Decision-Making and Planning for Intelligent Robotic Systems
- Multi-Agent Collaboration and Distributed Perception for Networked Autonomous Platforms

Research Assistant, Coordinated Systems Lab, Iowa State University (Aug 2023 – Jul 2025)

Advisor: Prof. Cody Fleming

- Safe Robot Learning and Safety-Constrained Decision-Making for Safety-Critical Systems
 - Offline Reinforcement Learning Algorithms and Data-Driven Control Methods
 - Learning-based Autonomous Driving/Racing Methods for F1tenth Robotic Vehicles
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PUBLICATIONS

Published/Accepted Works:

9. **Prajwal Koirala** and Mark Campbell. “Learning Communication-Conditioned Generative Policies for Decentralized Multi-Agent Collision Avoidance.” *International Conference on Intelligent Robots and Systems (IROS 2026)*.

8. **Prajwal Koirala** and Mark Campbell. “VGFM: Expressive Robot Policies via Dense Value Guidance in Flow Matching.” *International Conference on Intelligent Robots and Systems (IROS 2026)*.

7. **Prajwal Koirala** and Cody Fleming. “Flow-Based Single-Step Completion for Efficient and Expressive

- Policy Learning.” *International Conference on Learning Representations (ICLR 2026)*. [[Link](#)].
6. Nitesh Subedi, Adam Haroon, Samuel Tetteh, **Prajwal Koirala**, Cody Fleming, and Soumik Sarkar. “LCLA: Language-Conditioned Latent Alignment for Vision-Language Navigation.” *Conference on Robot Learning (CoRL 2026)*. [[Link](#)].
 5. Hsin-Jung Yang, Zhanhong Jiang, **Prajwal Koirala**, Qisai Liu, Cody Fleming, and Soumik Sarkar. “LexiSafe: Offline Safe Reinforcement Learning with Lexicographic Safety-Reward Hierarchy” *International Conference on Cyber-Physical Systems (ICCPs 2026)*. [[Link](#)]. **Best Paper Nominee**.
 4. **Prajwal Koirala**, Zhanhong Jiang, Soumik Sarkar, and Cody Fleming. “FAWAC: Feasibility Informed Advantage Weighted Regression for Persistent Safety in Offline Reinforcement Learning.” *Conference on Decision and Control (CDC 2025)*. [[Link](#)].
 3. **Prajwal Koirala**, Zhanhong Jiang, Soumik Sarkar, and Cody Fleming. “Latent Safety-Constrained Policy Approach for Safe Offline Reinforcement Learning.” *International Conference on Learning Representations (ICLR 2025)*. [[Link](#)].
 2. **Prajwal Koirala** and Cody Fleming. “Solving Offline Reinforcement Learning with Decision Tree Regression.” *Conference on Robot Learning (CoRL 2024)*. [[Link](#)].
 1. **Prajwal Koirala** and Cody Fleming. “Fltenth Autonomous Racing With Offline Reinforcement Learning Methods.” *IEEE Intelligent Transportation Systems Conference (ITSC 2024)*. [[Link](#)].

Peer-reviewed Workshop:

1. Nitesh Subedi, Adam Haroon, Shreyan Ganguly, Samuel Tetteh, **Prajwal Koirala**, Cody Fleming, and Soumik Sarkar. “Can Pretrained Vision-Language Embeddings Alone Guide Robot Navigation?” *RSS Workshop on Robot Planning in the Era of Foundation Models (RSS 2025: FM4RoboPlan)*. [[Link](#)].

Preprint / Under Review:

2. **Prajwal Koirala**, Mark Campbell. “REACH: Trust-Weighted Sensor Fusion through Reachability Estimation for Robust Robot Policy Learning.” (*Under Review*).
 1. Muhammad Bilal Shahid, Zhanhong Jiang, **Prajwal Koirala**, Soumik Sarkar, Cody Fleming. “Neural CDEs as Correctors for learned time series models.” (*Under Review*). [[Link](#)].
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EXPERIENCE

Department of Mechanical Engineering, Iowa State University, Ames, Iowa

Teaching Assistant

(Aug 2023 – Dec 2023)

- Led weekly lab sessions for ME421 *System Dynamics and Controls*
- Provided Matlab and Simulink training on system simulation and control system design

North Star Developer’s Village, Lalitpur, Nepal

R&D Engineer

(May 2022 – Jul 2023)

- Designed a customizable wheeled robot platform with ROS2-based navigation system
 - Enabled intuitive tracking/control with web visualization for enhanced robotics functionality
 - Worked with image/point-cloud data from LiDARs and depth cameras
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SKILLS

Programming Languages: Python, C, MATLAB, Javascript

CAD & Simulation: Autodesk Fusion 360, AutoCAD, ANSYS, Simulink

AWARDS AND RECOGNITION

Excellence Awards:

- Excellence in Research Award, **Translational AI Center, Iowa State University** (2025)
- Research Excellence Award, **Graduate College, Iowa State University** (2025)

Fellowship/Scholarship:

- Cornell PhD Fellowship, **Cornell University** (2025)
- Graduate Tuition Award for MS degree, **Iowa State University** (2023)
- Merit-based Scholarship for BE program at Pulchowk Campus, **Institute of Engineering, Nepal** (2017)
- Full Scholarship for studies in Science Faculty, **National Examination Board, Nepal** (2015)

Competitions:

- **Third Place**, CVPR 2025 AVA Workshop - 3D Human Motion Generation Challenge (2025)
- **Second Place**, Corn Yield Prediction Using Satellite Data, *6th Intl MLCAS Workshop*, (2024)
- **First Place**, Model-Based Design in MATLAB/Simulink, *DELTA 2.0*, Nepal (2021)
- **Second Place**, Jigyasa: National Engineering Expo, *Thapathali Campus, Nepal* (2019)

Academic Distinctions:

- **Highest Rank**, Mechanical Engineering Stream, *Institute of Engineering Entrance Examination* (2017)
 - **Highest Scorer**, Science Stream, *National Examination Board (Grade XI-XII), TTIC* (2017)
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ACADEMIC SERVICE AND ACTIVITIES

Reviewing:

Machine Learning: ICLR 2026, NeurIPS 2026, AAAI 2027, ICLR 2027

Robotics and Control: L4DC 2024, IROS 2026, ICRA 2027

Intelligent Transportation & Connected Systems: IEEE ITSC 2024, IEEE IV 2024, IEEE TMC (2x)

Computer Vision: WDFM-AD (ICCV 2025)

Mentoring:

- Mentored 3 summer undergrad interns over 2 years as part of *Research Experience for Undergraduates in Translational Applications of AI to Engineering (REU TrACE)* (2024, 2025)
- Volunteered as a math tutor for middle school students in **CyMath**, Iowa State University (2025)

Invited Talks & Oral Presentations:

- *Attention-Based Corn Yield Prediction with Multi-Source and Multi-Temporal Satellite Imagery*, **MLCAS**, Lincoln, Nebraska (2024)
 - *Reframing Offline Reinforcement Learning as a Regression Problem*, **TrAC Flash Talk**, Ames, Iowa (2024)
 - *Offline Reinforcement Learning: Overview and Recent Advances*, **Guest Lecturer (ME 5920)**, Iowa State University, Ames, Iowa (2025)
 - *Text-Conditioned Image Inpainting for 3D Human Motion Generation via Flow Matching*, **CVPR AVA: Accessibility, Vision, and Autonomy Meet**, Nashville, Tennessee (2025)
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